MIL-I-14804(MU)
23 June 1969
SUPERSEDING
FA-PD-MI-2650, Rev. 1
14 June 1965

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED: DELAY M4A1, M12A1, M45A1 and M49A1 REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers remanufactured, Initiator, Cartridge Actuated, Delay, M4A1, M12A1, M45A1 and M49A1, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-1167	- Ammunition Data Card

DRAWINGS

Munitions Command

D8595817	- Initiator, Cartridge Actuated, Delay, Metal Parts Assembly
C8796826	- Container, Ammunition, Fiber, M375 for Initiator, Delay, M4, M6A1, M12, M26, M33, M45 or M49

FSC 1377

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2.1 (Cont'd)

DRAWINGS

C8796827	- Box, Packing, Ammunition, for Initiator Delay, M4, M6A1, M12, M26, M33, M45 or M49 in Fiber Container, M375
IEL 8595817	- Inspection Equipment List for Initiator, Cartridge Actuated, M4A1, M12A1, M45A1 and M49A1, Metal Parts Assembly

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained by the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture, Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Hydrostatic pressure.- The initiator cap, chamber and plug simulating the firing pin housing, without firing pin hole and with an O-ring shall withstand a hydrostatic pressure of 5500 to 6500 pound per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.2).

3.3 Firing pin release.- The firing pin of the assembled initiator, less safety pin, cartridge retainer and chamber, shall release from the cocked position when the application of an axial load of 20 to 35 pounds has withdrawn the initiator pin 3/4 inches, maximum.

3.4 Initiator pin security.- The initiator pin of the assembled initiator, less cartridge retainer and chamber, shall not withdraw beyond the safety pin when subjected to an axial laod of 200 to 220 pounds. Application of the load shall cause no deformation of the safety pin.

3.5 Metal parts assembly.- All threaded joints shall be assembled hand tight. The remanufactured initiator shall contain a new (1) initiator spring (2) safety pin (3) O-ring and seal.

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3.6 Protective finish.- Protective finish shall be in accordance with the applicable drawings and referenced specifications and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish applied. Exterior remanufactured components shall consist of (1) initiator pin, (2) cap and (3) chamber.

3.7 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of 6 inches. The residual magnetism test may be waived on the metal parts assembly provided it is performed on the loaded assembly in an integrated plant.

3.8 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, reference specifications and the following:

3.8.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.8.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

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4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.- A lot of metal parts assemblies shall consist of 600 initiators, maximum, exclusive of test samples, provided the homogeneity requirements of the remanufacturing process have been retained in the inspection lots.

4.2.2 Examination.- The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for each defect as burr and foreign matter. Examination of reusable parts which require no processing shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All non-conforming parts shall be rejected. One hundred percent examination shall be performed for all critical defects on reprocessed parts. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables I and II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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Table II

Sample Sizes for Major Defects Number of defects in class	Total Allowable Defects Per Sample Size 110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

Sample Sizes for Minor Defects Number of defects in class	Total Allowable Defects Per Sample Size 110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.2.2.1 Pin, Initiator, Dwg. C8595403, covering a detail of Dwg. D8595817.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish on body and body groove improper	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Surface finish improper, except as otherwise classified	Visual

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4.2.2.2 Cap, Drawing D8595404, covering a detail of Dwg. D8595817.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of large thread, max.	Gage
102. Minor diameter of large thread, max.	Gage
103. Pitch diameter of small thread, max.	Gage
104. Minor diameter of small thread, max.	Gage
105. Thread damaged	Visual
106. Metal defective, except as otherwise classified	Visual
107. Protective finish, inadequate or defective	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
None defined	

4.2.2.3 Housing, Firing Pin, Dwg. D8595408, covering a detail of Dwg. D8595817.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of small thread, min.	Gage
102. Major diameter of small thread, min.	Gage
103. Pitch diameter of large thread, min.	Gage
104. Major diameter of large thread, min.	Gage
105. Thread damaged	Visual
106. Protective finish inadequate or defective	Visual
107. Surface finish in cavity improper	Visual
108. Burr or sharp corners	Visual
109. Foreign matter	Visual
MINOR	
201. Surface finish improper, except as otherwise classified	Visual

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4.2.2.4 Pin, Firing, Dwg. C8595407, covering a detail of Dwg. D8595817.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Surface finish improper	Visual
104. Burr or sharp edge	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Chamber, Initiator, Dwg. D8593314, covering a detail of Drawing D8595817.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Pitch diameter of large thread, min.	Gage
102. Major diameter of large thread, min.	Gage
103. Pitch diameter of small thread, min.	Gage
104. Major diameter of small thread, min.	Gage
105. Thread damaged	Visual
106. Protective finish inadequate or defective	Visual
107. Surface finish, end of gas port, improper	Visual
108. Burr	Visual
109. Foreign matter	Visual
MINOR	
201. Surface finish improper, except as otherwise classified	Visual

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4.2.2.6 Retainer, Cartridge, Dwg. D8593313, covering a detail of Dwg. D8595817.

Defect	Method of Inspection
CRITICAL	
1. Metering slots in base missing	Visual
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Elongated slots missing	Visual
104. Thread damaged	Visual
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Assembly, Dwg. D8595817, less all drawings except D8595408, C8595407, C8595403, B8595406, CCAX3-587929 and CKCX2-501462.

CRITICAL	
1. Protrusion of firing pin	Gage
2. Steel ball missing	Gage
3. "O" ring missing	Visual
MAJOR	
None defined	
MINOR	
201. Lubricant missing from "O" ring	Visual

4.2.2.8 Assembly, Dwg. D8595817.

CRITICAL

None defined

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4.2.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Initiator pin not free to rotate	Manual
102. Safety pin loose or deformed	Manual & Visual
103. Any part missing	Visual
104. Improper assembly	Visual
MINOR	
201. Shipping cap missing	Visual
202. Protective finish inadequate or defective	Visual

4.2.2.9 Packing in Fiber Container, Dwg. C8796826.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Gap between body and cover, max.	Manual & Visual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.10 Packing in Wooden Box, Dwg. C8796827.

CRITICAL

None defined

MAJOR

None defined

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4.2.2.10 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Hardware missing,loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each unitiator chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirements of 3.2 shall be cause for rejection of the subassembly.

4.2.3.2 Firing pin release.- The initiator pin of each subassembly shall be subjected to the firing pin release test. Failure of any subassembly to comply with the requirements of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Initiator pin security.- The initiator pin of each subassembly shall be subjected to initiator pin security test. Failure of any subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

4.2.3.4 Protective finish.- The number of specimens and acceptance criteria for protective finish shall be in accordance with the applicable Military Standard listed on the drawing.

4.2.3.5 Residual magnetism.- Each initiator shall be tested for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.7 shall be cause for rejection of the initiator.

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4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 8595817.

4.3 Test method and procedure.

4.3.1 Hydrostatic pressure.- The subassembly shall be mounted in a device using the fixture for properly sealing all openings, and subjected to the hydrostatic pressure to determine compliance with the requirements of 3.2.

4.3.2 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull.

4.3.3 Initiator pin security.- The subassembly shall be secured in the fixture with the initiator pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in each position.

4.3.4 Protective finish.- Test procedures for protective finish or re-plated components shall be in accordance with the applicable military standards referenced on the drawing.

4.3.5 Residual magnetism.- The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly 6 inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved slowly past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with drawings C8796826 and C8796827.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement document should specify the title, number and date of this specification.

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6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2650 dated 14 June 1965.

Custodian:

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Preparing activity:

Army - MU

Project No. 1377-A271

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